Supplementary Material for "CoSI: A Fine-grained Multimodal Dataset for Collaborative Social Interaction Understanding"

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1 Dataset Access

The CoSI dataset and benchmark resources are available at:

https://github.com/kenian123/CoSI

CoSI is a fine-grained multimodal dataset for collaborative social interaction understanding in small-group learning scenarios. The current release contains 10 video clips collected from 3 student groups with 9 participants in total. After temporal segmentation and annotation, the dataset contains 2,503 annotated interaction instances. If temporal windows are deduplicated by clip identifier and time span, the release contains 2,260 unique 5-second windows.

Each interaction instance is temporally aligned across six annotation dimensions: speaker, addressee, competency category, gesture, gaze, and body orientation. Excluding speaker identity, the annotation schema contains 17 fine-grained category labels. The dataset is organized for benchmark reuse in multimodal social interaction research, including participant-centered interaction understanding, higher-level collaborative semantics, and nonverbal cue recognition.

The repository is the official access point for the dataset release and benchmark resources associated with the paper. If needed for the final ACM MM submission, the following metadata can be added to this section in camera-ready form without changing the rest of the supplement: license, terms of use, citation file, and long-term archival mirror.

1.1 License and Access

The CoSI dataset is publicly accessible via the official dataset release link:

https://drive.google.com/drive/folders/1-wZWaej0gA5TBURfj89NWzu0vzd5NCr1?usp=sharing

The public release provides the benchmark resources and annotation files associated with this paper, enabling reproducible research on collaborative social interaction understanding. The dataset is intended for academic and research use in areas such as multimodal learning, educational multimedia analysis, and small-group collaboration modeling.

All participants provided informed consent for data collection and research use before recording. The release of the dataset is conducted on the basis of this informed consent. Users should handle the data responsibly and cite the associated CoSI paper when using the dataset in published work.

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2 Annotation Protocol

2.1 Data Source and Segmentation

CoSI is derived from a technology-enhanced engineering learning course on 3D modeling and printing. The original corpus contains approximately 200 hours of classroom recordings. From this corpus, we selected discussion-rich collaborative sessions in which each group worked for around one hour on authentic course-related tasks, including solution discussion, modeling reasoning, task coordination, resource allocation, and peer communication.

Audio and video were recorded in a smart classroom environment. Video was captured with a Hikvision AX-C22P+ device and audio was recorded with an iFLYTEK B1-32G recorder. The research team temporally aligned and merged the audio-visual streams, and then completed transcription, segmentation, and annotation in ELAN.

To construct a benchmark suitable for fine-grained social interaction analysis, the synchronized recordings were segmented into fixed 5-second windows. These windows were then converted into annotated interaction instances. Because more than one participant-centered interaction may occur within the same 5-second span, a single temporal window may correspond to multiple annotated instances.

2.2 Annotation Dimensions

Each interaction instance is labeled along six dimensions: Speaker, Competency, Addressee, Gesture, Gaze, and Body. To remain fully consistent with the label distribution shown in the main paper, we explain only the labels explicitly presented in the figure, namely `Spk-1/2/3`, `Comp-A/B/C/D`, `Addr-1/2/3`, `Gest-A/B/C`, `Gaze-A/B/D`, and `Body-A`. In particular, we do not expand `Gaze-C` or `Body-B/C` in this supplementary document.

This annotation design jointly captures verbal, interpersonal, and embodied social signals within the same local interaction window, making the dataset suitable for both recognition and higher-level reasoning tasks.

2.3 Annotation Workflow

To ensure annotation quality, we adopted a four-stage workflow:

(1) **Slicing:** aligned classroom video and separately recorded audio were merged and segmented into fixed 5-second windows in ELAN.
(2) **Initial annotation:** annotators labeled speaker, addressee, competency category, gesture, gaze, and body orientation for each window according to a unified coding manual.
(3) **Cross-checking:** annotations were reviewed to identify temporal misalignment, logical inconsistency across multimodal labels, and annotation drift across groups.
(4) **Structured integration:** reviewed annotations were cleaned and consolidated into the final benchmark-ready release.

Table 1: Definitions of the primary annotation labels used in the current paper. To keep the supplementary material fully consistent with the label distribution shown in the main paper, we report only the labels explicitly presented in the figure.

Dimension	Label	Full Name	Definition
Speaker	Spk-1	Speaker 1	The first participant in the group who is identified as the active speaker in the current interaction window.
Speaker	Spk-2	Speaker 2	The second participant in the group who is identified as the active speaker in the current interaction window.
Speaker	Spk-3	Speaker 3	The third participant in the group who is identified as the active speaker in the current interaction window.
Competency	Comp-A	Engineering Thinking and Problem Solving	This category corresponds to the first-level indicator A, *Engineering Thinking & Problem-Solving*. Its core emphasis is the systematic ability to transform abstract ideas into concrete and feasible solutions in collaborative engineering tasks.
Competency	Comp-B	Technical and Procedural Ability	This category corresponds to the first-level indicator B, *Technical & Process Proficiency*. It emphasizes practical skills for operating 3D-printing equipment and related software/hardware safely, efficiently, and with high quality.
Competency	Comp-C	Interdisciplinary Knowledge Integration	This category corresponds to the first-level indicator C, *Cross-Disciplinary Knowledge & Integration*. It reflects the ability to understand the scientific principles behind 3D printing and integrate them with knowledge from other disciplines.
Competency	Comp-D	Professionalism and Collaboration	This category corresponds to the first-level indicator D, *Professionalism & Collaboration*. It captures responsibility, teamwork, and communication ability demonstrated during project-based collaborative work.
Addressee	Addr-1	Addressee 1	The first participant in the group who is identified as the target addressee of the current speaker.
Addressee	Addr-2	Addressee 2	The second participant in the group who is identified as the target addressee of the current speaker.
Addressee	Addr-3	Addressee 3	The third participant in the group who is identified as the target addressee of the current speaker.

Table 2: Definitions of the embodied cue labels used in the current paper.

Dimension	Label	Full Name	Definition
Gesture	Gest-A	Deictic Gesture	A pointing or indicating gesture, such as pointing to a screen, model, or other task-relevant object.
Gesture	Gest-B	Representational Gesture	A gesture used to depict shapes, structures, spatial relations, or other representational content during discussion.
Gesture	Gest-C	Operational Gesture	A task-oriented action gesture, such as manipulating materials, disassembling parts, measuring data, or drawing during collaborative work.
Gaze	Gaze-A	Looking at Peers	The participant's gaze is directed toward other group members during the interaction.
Gaze	Gaze-B	Looking at Notebook or Tablet	The participant's gaze is directed toward a notebook, tablet, or similar learning aid.
Gaze	Gaze-D	Looking at Other Task-Relevant Objects	The participant's gaze is directed toward other relevant physical objects, such as models or materials used in the collaborative task.
Body	Body-A	Oriented Toward the Group	The participant's body orientation is aligned with and engaged in the small-group interaction.

The resulting annotations are intentionally information-dense. Rather than emphasizing large-scale volume, CoSI focuses on synchronized multimodal evidence and fine-grained labels that support analysis from individual behavior to interpersonal coordination and group-level collaboration patterns.

3 Agreement Details

To quantify annotation consistency, four annotators independently labeled a 15-minute classroom video segment. Inter-annotator agreement was measured using Fleiss' kappa, and the overall Fleiss' kappa is 0.8390, indicating strong agreement.

Agreement was evaluated after annotators followed the same coding manual and labeling procedure used for the main dataset release. The cross-checking stage focused on three common sources of disagreement: temporal boundary offset, inconsistency between verbal and nonverbal labels within the same window, and annotation drift across groups or sessions. The final release was obtained after structured cleaning and consolidation.

One distinctive property of CoSI is that the benchmark unit is interaction-centered rather than window-centered. Multiple participant-centered interactions may occur in the same 5-second span, so one temporal window can map to multiple final instances. This many-to-one relation increases annotation density and should be taken into account when interpreting agreement and benchmark difficulty.

4 Benchmark Details

4.1 Supported Task Family

Each annotated interaction instance defines a benchmark unit consisting of a 5-second multimodal segment and its aligned social labels. Based on the six annotation dimensions, CoSI supports a family of benchmark tasks:

- speaker recognition,
- addressee recognition,
- competency category prediction,
- gesture recognition,
- gaze prediction, and
- body-orientation recognition.

Together, these tasks span individual behavior recognition, interpersonal coordination modeling, and higher-level collaborative understanding.

4.2 Representative Benchmark: Competency Category Prediction

In the main paper, we use competency category prediction as the representative benchmark because it directly tests whether a model can connect multimodal interaction cues to higher-level collaborative meaning. We formulate a binary classification task that predicts whether competency category D (*professionalism and collaboration*) is present in a given annotated interaction instance.

For each instance, the model input consists of the aligned 5-second audio segment together with the corresponding middle video frame. All methods are evaluated on the same set of 2,503 annotated interaction instances using leave-one-video-out cross-validation. We report Accuracy (Acc), Balanced Accuracy (Bal Acc), Precision (Prec), Recall (Rec), F1 score (F1), and Area Under the ROC Curve (AUC).

We compare handcrafted baselines and frozen pretrained representation models. The handcrafted baselines include audio-only, video-only, and multimodal variants. The pretrained encoders include Wav2Vec2 Base for audio, and Swin V2-B, VideoMAE V2 (ViT-B), and DINOv2 ViT-B/14 for visual input.

4.3 Implementation Environment

All benchmark experiments were run with PyTorch 2.3.0 and Python 3.12 on Ubuntu 22.04, using CUDA 12.1. The experiments were conducted on a single NVIDIA V100 GPU with 32 GB memory. The CPU environment consisted of 6 vCPUs based on an Intel Xeon Gold 6130 processor running at 2.10 GHz. We report this configuration to facilitate reproducibility of the benchmark results.

For the pretrained-model baselines, we follow the evaluation setting described in the main paper: frozen pretrained encoders are used to extract representations from the aligned audio segment or visual input, and all methods are evaluated under the same leave-one-video-out protocol. The quantitative results are reported in the main paper. Overall, the benchmark demonstrates that CoSI remains challenging for generic pretrained representations, and that socially entangled collaboration signals are not captured well by off-the-shelf appearance features alone.

4.4 Additional Benchmark: Addressee Recognition

To demonstrate that CoSI also supports participant-centered interaction understanding, we additionally evaluate addressee recognition as a 3-class single-label classification task. Given a 5-second interaction instance, the goal is to predict whom the active speaker is addressing.

Following the same leave-one-video-out cross-validation protocol, we retain only samples with exactly one valid addressee label and accessible aligned media. Samples without a valid addressee label, with multiple positive addressee labels in the same window, or with failed media extraction are discarded. This yields 523 valid interaction instances from 10 videos, with class counts of 155, 221, and 147. The fold assignment is performed by clip identifier rather than by individual instances, ensuring that no video appears in both training and test folds.

For preprocessing, each retained interaction instance is converted into an addressee-recognition task record containing the clip identifier, group identifier, start time, end time, audio path, middle-frame path, short video-clip path, addressee label, and fold identifier. For the visual clip baseline, we uniformly sample 16 frames from the 5-second window. We report Accuracy, Balanced Accuracy, Macro-Precision, Macro-Recall, and Macro-F1.

We evaluate a majority-class predictor and three frozen pretrained baselines. Wav2Vec2 Base uses the 5-second audio segment as input, mean pools the last-layer hidden states, and applies a lightweight MLP classifier with dimensions $768 \rightarrow 256 \rightarrow 3$. DINOv2 ViT-B/14 uses the middle frame as input and feeds the pooled image representation into the same $768 \rightarrow 256 \rightarrow 3$ classifier. VideoMAE V2 (ViT-B) uses the 16-frame short clip as input and applies the same lightweight classifier to the pooled video representation. In all three baselines, the pretrained backbone is frozen and only the classification head is trained, keeping the benchmark lightweight and reproducible.

The benchmark results reported in the main paper show that addressee recognition also remains difficult under cross-video generalization, indicating that interaction targets depend on subtle and context-dependent multimodal cues.

5 Additional Examples

Table 3 provides several real annotation examples drawn from the release spreadsheets. To avoid overloading the table, we anonymize participants as P1, P2, and P3 within each group. These examples illustrate three important properties of CoSI: (1) multiple labels may co-occur in the same dimension, especially for addressee, competency category, and gaze; (2) multiple participant-centered interactions may be instantiated within the same time span; and (3) the dataset mixes higher-level collaborative semantics with low-level embodied cues in a temporally aligned format.

For consistency with the current paper, the supplementary document explains only the labels explicitly displayed in the main figure: `Comp-A/B/C/D`, `Gest-A/B/C`, `Gaze-A/B/D`, and `Body-A`. The mappings are summarized above so that future users can reproduce the reported benchmarks without ambiguity.

Table 3: Example annotated interaction instances from CoSI. "Comp." uses the rubric labels A/B/C/D defined in Section 2.2.

Group	Clip	Time (s)	Speaker	Addressee	Comp.	Gesture	Gaze / Body	Interpretation
G1	1	0–5	P2	P3	A, D	–	peers + notebook / toward group	A participant addresses one peer while jointly reasoning about the task and maintaining collaborative engagement.
G1	1	5–10	P1	P2, P3	D	deictic	peers / toward group	The active speaker addresses both teammates while pointing to a shared artifact, indicating coordinated group discussion.
G1	1	10–15	P3	P1, P2	B, D	–	peers / toward group	The speaker addresses both teammates in a procedural discussion while remaining embedded in the group interaction.
G2	1	5–10	P2	P1	D	deictic	other object / toward group	The speaker points to a task-relevant object or model while addressing a single teammate, reflecting localized collaboration around an artifact.
G2	1	15–20	P1	P2	A	–	peers / toward group	A focused dyadic exchange within the group illustrates that local interactions can be narrower than the full group context.